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I am pleased to formally introduce the **KiliKawi Challenge**: an innovation initiative supported by the Toyota Mobility Foundation and organized in partnership with East African Smart Villages & Rural Mobility (EASVRM).

The challenge is designed to catalyze innovation in developing next-generation electric agricultural tools and services for smallholder farmers in East Africa. As one of Kenya's leading technical universities, JKUAT is uniquely positioned to participate and contribute meaningfully to this initiative.

We would like to:

1. Invite JKUAT students to participate directly in the challenge.
2. For those with the strongest concepts and skills, be invited to our workshop, where they will receive mentorship and further support to refine and develop their prototypes.
3. Engage JKUAT students as local collaborators; working alongside participants from other parts of Africa.

This creates an excellent opportunity for your students to gain hands-on innovation and prototyping experience, collaborate internationally, and contribute to solutions with strong community impact.

## 1. Timeline

- End Oct 2025: Official Challenge Launch @ Nairobi Innovation Day
- 15 Jan 2026: Proposal submission deadline
- 15 Feb 2026: Announcement of top 5 student projects
- 01 Mar 2026: Kick-off workshop + mentor/team matchmaking
- Mar–May 2026: Project work phase with mentor check-ins
- End Jun 2026: Final presentations at showcase event (e.g., Agritec Africa)

## 2. Suggested Problem Areas

### *Potential Engineering Challenges*

- Design a docking system to improve PTO connection across appliances (miller, thresher, chaff cutter, etc.).
- Redesign the throttle system for single-person use during PTO operations.
- Develop a modular, lightweight standalone powertrain for multi-implement use.
- Prototype a hybrid power setup (battery + plug-in) for dual-mode operation.
- Integrate IoT for usage tracking, runtime, appliance recognition, and platform connection.

### *Potential Business Model & Economics Challenges*

- Propose appliance-specific invoicing models (e.g., per kg, per acre, per minute).
- Explore asset financing models (leasing, pay-as-you-go, subscription) for both agents and single farmers.

Please find more details on the challenge here: [KiliKawi Challenge Website](#).

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